

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459348.7423 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.047
Transit depth (Rp/Rs)^2	2.9 +/- 1.59 [%]
Orbital Inclination (inc)	87.53 +/- 2.06
Airmass coefficient 1 (a1)	0.991 +/- 0.023
Airmass coefficient 2 (a2)	0.003 +/- 0.013
Scatter in the residuals of the lightcurve fit is	2.36 %
Best Comparison Star	#3 - [156, 347]
Optimal Aperture	2.82
Optimal Annulus	11.28
Transit Duration (day)	0.079 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.047 vs 0.1394 (deviation 0.65σ)
Duration fit vs published	0.079 d vs 0.0983 d (deviation 1.76σ)
Mid-time O–C	-17.4 min
Depth SNR	3.6
Residual scatter	2.36 %

Light curve

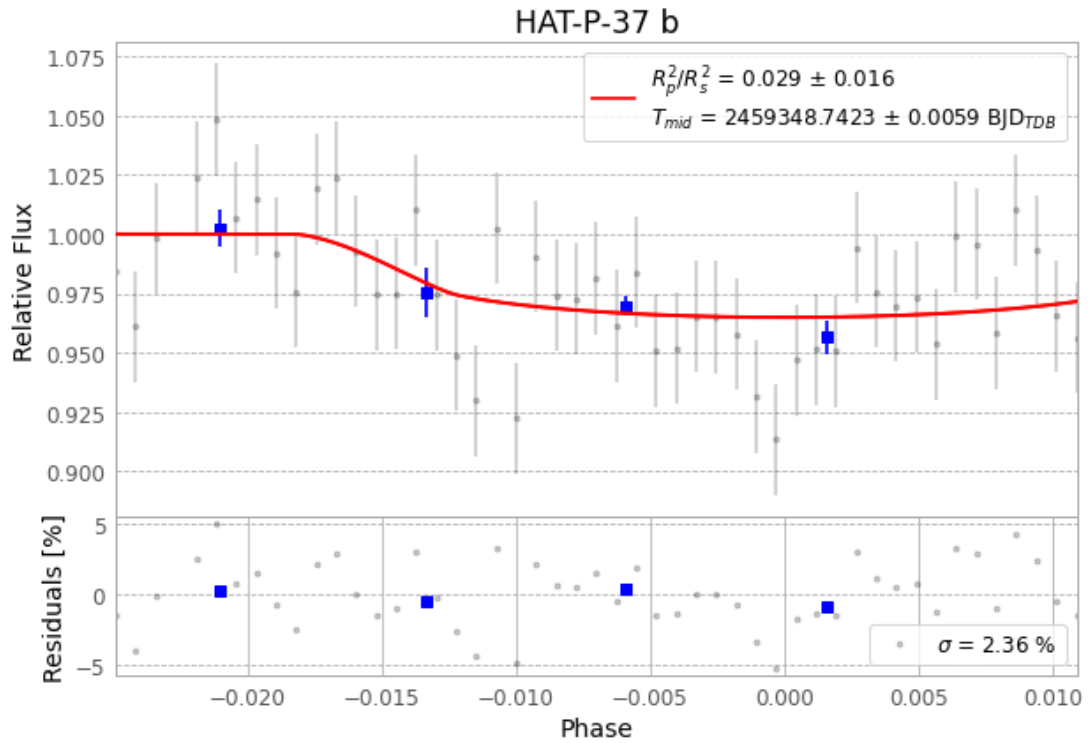


Figure 1 — FinalLightCurve_HAT-P-37 b_2021-05-13.png (SHA-256 70f844573eb23ab6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [227, 354], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f611bf84e4e3e81f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

49 FITS frames (32 MB), HATP-37210514040613.FITS → HATP-37210514063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-210514.log (992a091f2009...), FinalLightCurve_HAT-P-37 b_2021-05-13.png (70f844573eb2...), FinalLightCurve_HAT-P-37 b_2021-05-13.csv (d3d4248839b2...)

Compute resources

Wall time 97.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:53Z.